Jihyeok Park (박지혁)

Assistant Professor

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Research Interests

The ultimate goal is to make real-world software reliable and secure using programming language and software engineering techniques:

- **language specifications** to mechanize real-world specifications such as ECMA-262 and derive language tools from them.
- **program analysis** to statically and dynamically understand program behaviors and detect bugs.
- **automated testing** to generate tests, coverage criteria, and oracles for language implementations and specifications.
- **type systems** to guarantee program safety and precisely analyze programs and specifications.
- **program synthesis** to automatically generate programs from specifications and examples.
- **software security** to find and prevent vulnerabilities, from vulnerable JavaScript dependencies to memory-safety bugs.

Education

Integrated M.S. & Ph.D in School of Computing — *Daejeon, South Korea*

Korea Advanced Institute of Science and Technology (KAIST) — *Mar. 2016 - Feb. 2022*

- Ph.D. Thesis - JavaScript Static Analysis for Evolving Language Specifications
- Programming Language Research Group
- Advisor: Sukyoung Ryu

B.S in School of Computing and Mathematical Science — *Daejeon, South Korea*

Korea Advanced Institute of Science and Technology (KAIST) — *Mar. 2012 - Feb. 2016*

Work Experience

Mar. 2023 - PRESENT **Assistant Professor**, Korea University — *Seoul, South Korea*

Feb. 2022 - Feb. 2023 **Post Doctoral Fellow**, Oracle Labs — *Brisbane, Australia*

Publications

[TOSEM 2026] Detecting JavaScript Libraries on the Web using Property-Order Graphs

*Sungmin Park, Seojin Kim, Ben Swierzy, and **Jihyeok Park**†*

Transactions on Software Engineering and Methodology

[ASE 2026] MiniPoly: Automatic Extraction of Efficient JavaScript Polyfill from Language Specification

Jungwoong Kim, Youngmin Cho*, and **Jihyeok Park**† (* equally contributed)*

Proceedings of the IEEE/ACM International Conference on Automated Software Engineering

ACM SIGSOFT Distinguished Paper Award

[SAS 2026] Synthesizing Parsing Expression Grammars via Analysis-based Pruning

*Seongmin Ko and **Jihyeok Park**†*

Static Analysis Symposium

[TOSEM 2026] Selective Feature-Sensitive Coverage for Conformance Testing of Programming Languages

*Kanguk Lee, Seunghwan Kim, **Jihyeok Park**†, and Sukyoung Ryu†*

Transactions on Software Engineering and Methodology

[ASE 2025] Debun: Detecting Bundled JavaScript Libraries on Web using Property-Order Graphs

Seojin Kim, Sungmin Park*, and **Jihyeok Park**† (* equally contributed)*

Proceedings of the IEEE/ACM International Conference on Automated Software Engineering

[FSE 2025 Demo] JSSpecVis: A JavaScript Language Specification Visualization Tool

Minseok Choe, Kyungho Song*, Hyunjoon Kim, and **Jihyeok Park**† (* equally contributed)*

Proceedings of the ACM International Conference on the Foundations of Software Engineering Demonstration track

[PLDI 2024] PL4XGL: A Programming Language Approach to Explainable Graph Learning

Minseok Jeon, ***Jihyeok Park***†, *and Hakjoo Oh*†

Proceedings of the 45th ACM SIGPLAN Conference on Programming Language Design and Implementation

[CACM 2024] JavaScript Language Design and Implementation in Tandem

Sukyoung Ryu and ***Jihyeok Park***

Communications of the ACM

[PLDI 2023] Feature-Sensitive Coverage for Conformance Testing of Programming Language Implementations

Jihyeok Park, *Dongjun Youn*, *Kanguk Lee*, *and Sukyoung Ryu*

Proceedings of the 44th ACM SIGPLAN Conference on Programming Language Design and Implementation

[ESEC/FSE 2022] Automatically Deriving JavaScript Static Analyzers from Specifications using Meta-Level Static Analysis

Jihyeok Park, *Seungmin An*, *and Sukyoung Ryu*

Proceedings of the 30th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering

[PLDI 2022 Tutorial] Filling the Gap between the JavaScript Language Specification and Tools using the JISET Family

Sukyoung Ryu, ***Jihyeok Park***, *and Seungmin An*

Proceedings of the 43rd ACM SIGPLAN Conference on Programming Language Design and Implementation Tutorial

[ASE 2021] JSTAR: JavaScript Specification Type Analyzer using Refinement

Jihyeok Park, *Seungmin An*, *Wonho Shin*, *Yusung Sim*, *and Sukyoung Ryu*

Proceedings of the 36th IEEE/ACM International Conference on Automated Software Engineering

[ESEC/FSE 2021] Accelerating JavaScript Static Analysis via Dynamic Shortcuts

*Joonyoung Park**, ***Jihyeok Park****, *Dongjun Youn*, *and Sukyoung Ryu* (* *equally contributed*)

Proceedings of the 29th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering

[ECOOP/ISSTA 2021 DS] JavaScript Static Analysis with Evolving Engines and Specification

Jihyeok Park

Proceedings of the 35th European Conference on Object-Oriented Programming and Proceedings of the 30th ACM SIGSOFT International Symposium on Software Testing and Analysis Doctoral Symposium track

[CSUR 2021] A Survey of Parametric Static Analysis

Jihyeok Park*, *Hongki Lee**, *and Sukyoung Ryu* (* *equally contributed*)

ACM Computing Surveys, *Volume 54*, *Issue 7*, *Article No. 149*, *pp 1–37*

[ICSE 2021] JEST: N+1-version Differential Testing of Both JavaScript Engines and Specification

Jihyeok Park, *Seungmin An*, *Dongjun Youn*, *Gyeongwon Kim*, *and Sukyoung Ryu*

Proceedings of the 43rd ACM/IEEE International Conference on Software Engineering

ACM SIGSOFT Distinguished Paper Award

[ASE 2020] JISET: JavaScript IR-based Semantics Extraction Toolchain

Jihyeok Park, *Jihee Park*, *Seungmin An*, *and Sukyoung Ryu*

Proceedings of the 35th IEEE/ACM International Conference on Automated Software Engineering

[IEEE Software 2019] Towards Analysis and Bug Finding of JavaScript Web Applications in the Wild

Sukyoung Ryu, ***Jihyeok Park***, *and Joonyoung Park*

IEEE Software, *Volume 36*, *Issue 3*, *pp 74-82*

[Scala 2018] Path Dependent Types with Path-Equality

Jaemin Hong, ***Jihyeok Park***, *and Sukyoung Ryu*

Proceedings of the 9th ACM SIGPLAN Symposium on Scala

[ProWeb 2018] A Framework for Dynamic Inter-Device Task Dispatch with Eventual Consistency

Jihyeok Park, *Joonyoung Park*, *Yoonkyong Lee*, *Chul-Joo Kim*, *Byoungoh Kim*, *and Sukyoung Ryu*

Proceedings of the 2nd International Workshop on Programming Technology for the Future Web

[NetPL 2017] Toward Building Memory-safe Network Functions with Modest Performance Overhead

Keunhong Lee, *Shinae Woo*, *Sanghyeon Seo*, ***Jihyeok Park***, *Sukyoung Ryu*, *and Sue Moon*

Proceedings of the 3rd SIGCOMM Workshop on Networking and Programming Languages

[SOAP 2017] Revisiting Recency Abstraction for JavaScript: Towards an Intuitive, Compositional, and Efficient Heap Abstraction

Jihyeok Park, *Xavier Rival and Sukyoung Ryu*

Proceedings of the International Workshop on the State Of the Art in Java Program Analysis

Best Paper Award

[ICSE 2017 Demo] Analysis of JavaScript Web Applications Using SAFE 2.0

Jihyeok Park, *Yeonhee Ryou*, *Joonyoung Park*, *and Sukyoung Ryu*

Proceedings of the 39th International Conference on Software Engineering Demonstrations track

[Modularity 2014] JavaScript API Misuse Detection by Using TypeScript

Jihyeok Park

Proceedings of the 13th International Conference on Modularity

ACM Student Research Competition 3rd

Honors & Awards

Oct. 2026	**ACM SIGSOFT Distinguished Paper Award**, ASE 2026	*Munich*, *Germany*
Dec. 2023	**고려대학교 우수강좌**, Dept. Computer Science and Engineering, Korea University (COSE212)	*Seoul*, *South Korea*
Mar. 2022	**Ph.D. Dissertation Award**, College of Engineering, KAIST	*Daejeon*, *South Korea*
Feb. 2022	**An Outstanding Ph.D. Thesis**, School of Computing, KAIST	*Daejeon*, *South Korea*
May 2021	**ACM SIGSOFT Distinguished Paper Award**, ICSE 2021	*Madrid*, *Spain*
Dec. 2020	**PhD Fellowship Award**, NAVER Corp.	*Daejeon*, *South Korea*
Sep. 2019	**Outstanding Teaching Assistant Award**, School of Computing, KAIST (CS320)	*Daejeon*, *South Korea*
Feb. 2019	**Outstanding Teaching Assistant Award**, School of Computing, KAIST (CS320)	*Daejeon*, *South Korea*
Aug. 2018	**Outstanding Teaching Assistant Award**, School of Computing, KAIST (CS320)	*Daejeon*, *South Korea*
Aug. 2017	**Outstanding Teaching Assistant Award**, School of Computing, KAIST (CS320)	*Daejeon*, *South Korea*
Jun. 2017	**Best Paper**, International Workshop on the State Of the Art in Java Program Analysis (SOAP)	*Barcelona*, *Spain*
Feb. 2017	**Outstanding Teaching Assistant Award**, School of Computing, KAIST (CS320)	*Daejeon*, *South Korea*
Sep. 2016	**Outstanding Teaching Assistant Award**, School of Computing, KAIST (CS109)	*Daejeon*, *South Korea*
Feb. 2016	**Magna Cum Laude (GPA: 3.88/4.3)**, School of Computing, KAIST	*Daejeon*, *South Korea*
Sep. 2014	**Workshop 3rd Award**, URP Program, KAIST	*Daejeon*, *South Korea*
Apr. 2014	**ACM Student Research Competition 3rd**, International Conference on Modularity 2014	*Lugano*, *Swiss*
Mar. 2014	**Honor Program**, School of Computing, KAIST	*Daejeon*, *South Korea*
Jul. 2013	**KAIST Presidential Fellowship**, School of Computing, KAIST	*Daejeon*, *South Korea*

Teaching

Teacher

2026 Fall	**COSE212: Programming Languages**, Korea University	*Seoul*, *South Korea*
2026 Fall	**COSE405: Computer Science Colloquium**, Korea University	*Seoul*, *South Korea*
2026 Fall	**AAA705: Software Testing and Quality Assurance**, Korea University	*Seoul*, *South Korea*
2026 Fall	**SWS128: Software Analysis**, Korea University	*Seoul*, *South Korea*
2026 Spring	**COSE215: Theory of Computation**, Korea University	*Seoul*, *South Korea*
2026 Spring	**AAA551: Programming Language Theory**, Korea University	*Seoul*, *South Korea*
2025 Fall	**COSE212: Programming Languages**, Korea University	*Seoul*, *South Korea*
2025 Spring	**COSE215: Theory of Computation**, Korea University	*Seoul*, *South Korea*
2024 Fall	**COSE212: Programming Languages**, Korea University	*Seoul*, *South Korea*
2024 Spring	**COSE215: Theory of Computation**, Korea University	*Seoul*, *South Korea*
2024 Spring	**AAA705: Software Testing and Quality Assurance**, Korea University	*Seoul*, *South Korea*
2024 Spring	**SWS121: Secure Programming**, Korea University	*Seoul*, *South Korea*
2023 Fall	**COSE212: Programming Languages**, Korea University	*Seoul*, *South Korea*
2023 Spring	**COSE215: Theory of Computation**, Korea University	*Seoul*, *South Korea*

Teaching Assistant

2019 Fall	**CS320: Programming Languages course**, KAIST	*Daejeon*, *South Korea*
2019 Spring	**CS320: Programming Languages course**, KAIST	*Daejeon*, *South Korea*
2018 Fall	**CS408: Computer Science Project course**, KAIST	*Daejeon*, *South Korea*
2018 Fall	**CS320: Programming Languages course**, KAIST	*Daejeon*, *South Korea*
2018 Spring	**CS320: Programming Languages course**, KAIST	*Daejeon*, *South Korea*

2017 Fall	**CS492: Special Topics in Computer Science <Program Analysis>**, KAIST	*Daejeon, South Korea*
2017 Fall	**CS320: Programming Languages course**, KAIST	*Daejeon, South Korea*
2017 Spring	**CS320: Programming Languages course**, KAIST	*Daejeon, South Korea*
2016 Fall	**CS320: Programming Languages course**, KAIST	*Daejeon, South Korea*
2016 Spring	**CS109: Programming Practice course**, KAIST	*Daejeon, South Korea*

Talks

Oct, 2026	**UC Berkeley PLAIT Lab**, MiniPoly: Automatic Extraction of Efficient JavaScript Polyfill from Language Specification	*Berkeley, CA, USA*
Aug, 2025	**SIGPL Summer School 2025**, 안전한 자바스크립트 언어 생태계 확보 기술	*Seoul, South Korea*
Jul. 2025	**Programming Language Standardization and Specification**, Trusted JavaScript Language Environments with ESMeta	*Bergen, Norway*
Jun. 2025	**Real-World Programming Language Specification (RPLS)**, Trusted JavaScript Language Environments with ESMeta	*Seoul, South Korea*
May 2025	**고려대학교 CS 콜로퀴움**, 기계화 명세를 이용한 자바스크립트 언어의 설계와 구현	*Seoul, South Korea*
Apr. 2025	**Software Convergence Symposium 2025 (SWCS)**, 기계화 명세를 통한 안전한 자바스크립트 언어 생태계 확보 기술	*Seoul, South Korea*
Nov. 2024	**Ulsan National Institute of Science and Technology (UNIST)**, Language Design and Implementation using JavaScript Mechanized Specification	*Ulsan, South Korea*
Jun. 2024	**Instituto Superior Técnico of University of Lisbon**, Language Design and Implementation using JavaScript Mechanized Specification	*Lisbon, Portugal*
May 2024	**한국과학영재학교 수리정보과학부 콜로퀴움**, JavaScript 언어 생태계 자동화 연구를 하기까지의 여정	*Busan, South Korea*
Nov. 2023	**고려대학교 융합보안대학원 세미나**, 자바스크립트 엔진 보안과 퍼징	*Seoul, South Korea*
Oct. 2023	**고려대학교 CS 콜로퀴움**, 기계화 명세를 이용한 자바스크립트 언어의 설계와 구현	*Seoul, South Korea*
Aug. 2023	**SIGPL Summer School 2023**, PL 구현체를 위한 새로운 커버리지를 제안하기까지의 여정	*Chuncheon, South Korea*
May 2023	**고려대학교 고려대학교 KUGODS 연사 초청 특강**, 프로그래밍 언어 연구란?	*Seoul, South Korea*
Apr. 2023	**고려대학교 KU-KIST융합대학원 저널클럽**, Guiding the Design and Implementation of JavaScript using Mechanized Specification	*Seoul, South Korea*
Mar. 2023	**고려대학교 CS 콜로퀴움**, 기계화 명세를 이용한 자바스크립트 언어의 설계와 구현	*Seoul, South Korea*
Mar. 2023	**고려대학교 소프트웨어 분석 연구실 세미나**, 기계화 명세를 이용한 JavaScript 언어의 설계와 구현	*Seoul, South Korea*
Jan. 2023	**KCSE 2023**, Automatically Deriving JavaScript Static Analyzers from Specifications using Meta-Level Static Analysis	*Pyeongchang, South Korea*
Dec. 2022	**KSC 2022**, Automatically Deriving JavaScript Static Analyzers from Specifications using Meta-Level Static Analysis	*Jeju, South Korea*
Feb. 2022	**SIGPL Winter School 2022**, JavaScript Static Analysis for Evolving Language Specifications	*Online*
Feb. 2022	**STAAR Workshop**, JavaScript Static Analysis for Evolving Language Specifications	*Online*
Jan. 2022	**The 88th meeting of Ecma TC39**, JavaScript Static Analysis for Evolving Language Specifications	*Online*
Jan. 2022	**KCSE 2022**, JSTAR: JavaScript Specification Type Analyzer using Refinement	*Online*
Jan. 2022	**KCSE 2022**, JEST: N+1-version Differential Testing of Both JavaScript Engines and Specification	*Online*
Jan. 2022	**Agoric**, JavaScript Static Analysis for Evolving Language Specifications	*Online*
Dec. 2021	**KSC 2021**, JEST: N+1-version Differential Testing of Both JavaScript Engines and Specification	*Online*
Dec. 2021	**Département d'Informatique de l'École Normale Supérieure**, JavaScript Static Analysis for Evolving Language Specifications	*Online*
Oct. 2021	**EIRIC**, JEST: N+1-version Differential Testing of Both JavaScript Engines and Specification	*Online*
Sep. 2021	**KAISTPL Workshop 2021**, JavaScript Static Analysis for Evolving Language Specifications	*Online*
Jun. 2021	**KAIST Prosys Lab**, Towards Co-evolution of JavaScript Specification and Tools	*Daejeon, South Korea*
Jun. 2021	**KCC 2021**, JISET: JavaScript IR-based Semantics Extraction Toolchain	*Jeju, South Korea*
Nov. 2019	**National University of Singapore**, Update-Tolerant JavaScript Static Analysis for Frequently Released ECMAScript	*Singapore*

Software

ESMeta: ECMAScript Specification (ECMA-262) Metalanguage

Programming Language Research Group, KAIST — *Jan. 2022 - PRESENT*

- Main developer of ESMeta
- Developed in Scala
- URL: https://github.com/es-meta/esmeta

SAFE: Scalable Analysis Framework for ECMAScript

Programming Language Research Group, KAIST — *Mar. 2016 - Mar. 2021*

- Main developer of SAFE version 2.0
- Developed in Scala
- URL: https://github.com/sukyoung/safe

Activities

Organizing Committee

Jun. 2027	**SOAP 2027**, Program Co-Chair	*Atlanta*, *United States*
Jun. 2025	**RPLS 2025**, Organizing Committee Member	*Seoul*, *South Korea*
Jul. 2022	**ISSTA 2022**, Web Co-Chair	*Daejeon*, *South Korea*

Committee Members

Jul. 2027	**FSE 2027**, Program Committee Member	*Shenzhen*, *China*
Apr. 2027	**ICSE 2027**, Program Committee Member	*Dublin*, *Ireland*
Nov. 2026	**CCS 2026**, Program Committee Member	*Hague*, *Netherlands*
Oct. 2026	**ASE 2026**, Program Committee Member	*Munich*, *Germany*
Sep. 2026	**ICSME 2026**, Program Committee Member	*Benevento*, *Italy*
Jun. 2026	**SOAP 2026**, Program Committee Member	*Boulder*, *United States*
Jun. 2026	**PLDI 2026**, Review Committee Member	*Boulder*, *United States*
Mar. 2026	**SAC 2026 (PL Track)**, Program Committee Member	*Thessaloniki*, *Greece*
Oct. 2025	**OOPSLA 2025**, Review Committee Member	*Singapore*
Oct. 2025	**SAS 2025**, Program Committee Member	*Singapore*
Sep. 2025	**SCAM 2025**, Program Committee Member	*Auckland*, *New Zealand*
Apr. 2025	**SAC 2025 (PL Track)**, Program Committee Member	*Sicily*, *Italy*
Oct. 2024	**APLAS 2024 Student Research Competition & Posters**, Selection Committee Member	*Kyoto*, *Japan*
Oct. 2024	**SPLASH 2024 Doctoral Symposium)**, Program Committee Member	*Pasadena*, *United States*
Apr. 2024	**SAC 2024 (PL Track)**, Program Committee Member	*Avila*, *Spain*
Apr. 2024	**ICSE 2024**, Program Committee Member	*Lisbon*, *Portugal*
Dec. 2023	**OOPSLA 2023**, Extended Review Committee Member	*Lisbon*, *Portugal*
Nov. 2023	**APLAS 2023**, Program Committee Member	*Taipei*, *Taiwan*
Dec. 2022	**OOPSLA 2022**, Extended Review Committee Member	*Auckland*, *New Zealand*
Dec. 2022	**APLAS 2022**, Program Committee Member	*Auckland*, *New Zealand*

Artifact Evaluation Committee Members

Dec. 2023	**OOPSLA 2023**, Artifact Evaluation Committee Member	*Lisbon*, *Portugal*
Dec. 2022	**OOPSLA 2022**, Artifact Evaluation Committee Member	*Auckland*, *New Zealand*
Oct. 2019	**POPL 2019**, Artifact Evaluation Committee Member	*Louisiana*, *United States*
Jul. 2019	**OOPSLA 2019**, Artifact Evaluation Committee Member	*Athens*, *Greece*
Jul. 2018	**OOPSLA 2018**, Artifact Evaluation Committee Member	*Massachusetts*, *United States*
Jun. 2018	**SAS 2018**, Artifact Evaluation Committee Member	*Freiburg im Breisgau*, *Germany*
May. 2018	**ISSTA 2018**, Artifact Evaluation Committee Member	*Amsterdam*, *Netherlands*

Journal Reviewers

2026	**Transactions on Software Engineering and Methodology (TOSEM)**, Reviewer
2025	**Transactions on Software Engineering and Methodology (TOSEM)**, Reviewer
2024	**IEEE Transactions on Software Engineering (TSE)**, Reviewer

2023	**IEEE Transactions on Software Engineering (TSE)**, Reviewer	
2022	**IEEE Transactions on Software Engineering (TSE)**, Reviewer	

Others

Sep. 2015 - Feb. 2019	**Samsung Electronics**, Reviewer for C++ Code Reviewing Exams	*Daejeon*, *South Korea*
Jul. 2018	**École normale supérieure**, Internship Program (Professor: Xavier Rival)	*Paris*, *France*
Sep. 2017	**NII Shonan Meeting**, Topic: Memory Abstraction	*Shonan*, *Japan*
Feb. 2015 - Jun. 2015	**INSA de Toulouse**, Exchange Student Program	*Toulouse*, *France*
Jun. 2014 - Nov. 2014	**KOFAC**, Undergraduate Research Program (URP) for Fusion of Creatives	*Daejeon*, *South Korea*
Dec. 2013 - Jun. 2014	**School of Computing, KAIST**, Undergraduate Research Project (URP) Program	*Daejeon*, *South Korea*